

Jeremy Burke

Manchester, CT
(860) 808 - 7893
jeremy.burke024@gmail.com

Professional Summary

Applied AI & R&D Engineer delivering SBIR/STTR defense programs, specializing in computer vision, sensor fusion, and simulation across \$100K-\$1M+ efforts.

Experience

Computer Engineer — Beacon Industries — Newington, CT Aug 2025 – Present

- Led SBIR/STTR programs (DoD, NASA, MDA), delivering technical solutions and reports valued at \$30K-\$250K+ independently
- Developed and trained computer vision models and synthetic sensor pipelines for perception and fusion validation
- Engineered custom plugins for thermal and multi-spectral sensor simulation from raw data streams
- Built AI tools using AWS (S3, EC2), Weaviate, and OpenAI APIs for internal automation and decision support
- Presented technical work in reports and commercialization opportunities directly to government stakeholders

Selected Projects

Multi-Spectral Passive Object Detection (Navy SBIR)

- Built ML-based object detection pipeline using synthetic + real multi-spectral data
- Developed 3D physics simulations and sensor plugins to model environmental effects and enable sensor fusion testing

Hypersonic Kill Vehicle Research (MDA SBIR)

- Performed ANSYS simulations to evaluate high-speed system performance and design tradeoffs

Additive Manufacturing R&D (SBIR Phase II)

- Delivered \$500K in technical milestones and coordinated external research collaboration

Reverse Engineering (Defense System)

- Leading vendor coordination and technical analysis for legacy electromechanical system reconstruction

Education

University of Connecticut — Storrs, CT Aug 2019 – Dec 2023
B.S. Computer Science & Engineering, Minor Mathematics GPA: 3.3

Technical Skills

Programming: Python, C, C++, Java, SQL
AI/ML: Computer Vision, YOLOv8, Multimodal Fusion, LLM Systems
Simulation: Gazebo, ANSYS Workbench, Physics-Based Modeling
Cloud: AWS (S3, EC2), Weaviate, OpenAI API
Tools: MATLAB, LTspice, Wireshark, Linux, LaTeX